

Cat Lab

Design Report

Full-site audit — Home, Cat History, Gallery, Favourites

March 2026

Executive Summary

Cat Lab is a four-page vanilla HTML/CSS/JS website for browsing live cat photos, exploring cat history, and saving favourites. The overall design is warm, playful, and cohesive. The coral/cream/teal palette, Playfair Display + Inter typographic pairing, and CSS custom property system create a polished, consistent aesthetic. The most significant improvement areas are accessibility (color contrast, focus states) and a few consistency gaps across pages.

Five priority fixes have already been implemented:

- Color contrast — switched coral and teal text labels to their darker variants to meet WCAG AA
- Focus-visible styles — added keyboard focus outlines for all interactive elements
- Active nav state — current page is now highlighted in the navigation on all 4 pages
- Unified hero palette — history page (was dark brown) and favourites page (was purple) now use teal and coral respectively
- Touch targets — heart and wand buttons increased from 38 px to 44 px

Design System

Color Palette

Token	Hex	Use
--coral	#E8614B	Primary gradient backgrounds, interactive fills
--coral-dark	#C94A36	Text labels on light backgrounds (WCAG AA)
--coral-light	#F5907C	Gradient endpoints
--teal	#2A9D8F	Secondary actions, gallery hero, timeline
--teal-dark	#1D7A6F	Text labels, hover states
--cream	#FFF8F0	Page background
--cream-dark	#F5EDE0	Section bands, pill label backgrounds
--charcoal	#2C2C2C	Headings, footer background

--mid-grey	#6B6B6B	Body text, captions
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Typography

- **Display:** Playfair Display (serif) — h1 through h4, logo
- **Body:** Inter (sans-serif) — all UI text, buttons, captions
- **Scale:** Fluid using clamp() — h1 scales from 2.2 rem to 3.8 rem; no fixed breakpoint jumps

Spacing & Shape

- Border radius system: 8 px (sm) / 16 px (md) / 24 px (lg) / pill (9999 px)
- Shadow depth: sm (subtle card) / md (hover state) / lg (hero images, toasts)
- Consistent hover pattern: translateY lift + shadow step up

First Impression

The site communicates its purpose instantly. The coral hero on the home page, the cat emoji logo, and the 'Where Every Cat Tells a Story' headline establish personality within two seconds. The visual weight is well-calibrated — the CTA buttons are easy to find, and the grid of circular cat photos adds life to the hero without competing with the headline.

The history page timeline is the design standout: alternating left/right cards with scroll-triggered fade-in and a coral-to-teal gradient spine is the most sophisticated interaction on the site.

Usability Findings

Finding	Severity	Status	Recommendation
No active nav state on current page	Moderate	Fixed	Active class added to all 4 pages
Gallery action buttons below minimum touch target	Moderate	Fixed	Buttons increased from 38 px to 44 px
No keyboard focus indicator	Critical	Fixed	:focus-visible outlines added globally
No empty state for breed filter zero results	Moderate	Open	Add a friendly 'No cats found' message
Hero cat images have no loading placeholder	Minor	Open	Add skeleton shimmer circles while API images load
Hamburger menu missing aria-expanded	Moderate	Open	Toggle aria-expanded when menu opens/closes
No skip-to-content link	Minor	Open	Add a visually-hidden skip link for screen reader users

Visual Hierarchy

Home Page

- Reading order: Hero → Stats band → Feature cards → Showcase sections → CTA band — logical, guides the user toward the gallery
- The stats band (10,000+, 12, 1) is visually understated; numbers feel decorative without context labels
- Alternating showcase sections (text + image, flipped) work well for rhythm

Gallery Page

- Cards with min-height placeholders prevent layout shift when AI names load — good
- Empty space before AI naming is slightly awkward; consider a subtle 'tap to name' nudge label

History Page — standout

- Strongest visual storytelling on the site
- Coral/teal alternating dots mirror card borders — excellent system thinking
- Era labels (small uppercase) provide good information hierarchy within each card

Favourites Page

- Previously purple hero felt disconnected from brand; now uses coral gradient
- Empty state (paw emoji + CTA to gallery) is well-designed and friendly

Consistency Findings

Element	Issue	Recommendation
Feature card hover	translateY(-6px) vs cat card -4px/-5px	Standardise on -4px across all cards
Hero gradients	Now unified (coral = home/favourites, teal = history/gallery)	Maintain pattern going forward
Accent colors	Orange and purple appear only on feature cards	Replace with coral-light and teal-light to stay within 2-color system
btn-lg usage	Used inconsistently across sections	Reserve btn-lg for hero and section CTA bands only
Page-level CSS	Hero styles in inline <style> blocks in each HTML file	Move to style.css under named section comments

Accessibility Audit

Check	Result	Notes
Coral (#E8614B) on white	~3.4:1 FAIL AA	Fixed: --coral-dark (#C94A36) ~4.8:1 now used for text
Teal (#2A9D8F) on white	~3.7:1 FAIL AA	Fixed: --teal-dark (#1D7A6F) now used for text labels
Charcoal on cream	~13.5:1 PASS AAA	Headings and body text: excellent
White on coral gradient	~4.6:1 PASS AA	Hero text passes
Focus indicators	Missing FAIL	Fixed: :focus-visible outlines added globally
Touch target size	38 px FAIL	Fixed: increased to 44 px

Decorative emoji aria-hidden	Not set	Add aria-hidden='true' to all decorative emoji spans
Gallery image alt text	Partial	Update gallery.js to inject AI name into img.alt when assigned
Semantic landmarks	PASS	Nav, main, footer present — good structural HTML
Hamburger aria-expanded	Static 'false'	Toggle in app.js on menu open/close

Mobile & Responsive Review

The site uses a single breakpoint at 768 px. This is functional but leaves a gap for tablet-sized viewports.

- **769 px–1024 px (tablet):** Content is in full desktop layout but the viewport is narrower — 2-column showcase sections and 340 px timeline cards feel tight
- **Below 768 px:** Gallery grid drops to 160 px minimum columns; card content (AI name, personality, action buttons) is cramped — 200 px minimum recommended
- The sticky filter bar on the gallery page works well on mobile; full-width select is a good adaptive choice

Recommended: add a tablet breakpoint at 1024 px

- Showcase sections: switch to single column
- Timeline: reduce card max-width to 280 px
- Gallery grid: minmax(200 px, 1fr)

What Works Exceptionally Well

- **Design token system** — CSS custom properties for color, radius, shadow, and transition are centralised and used consistently throughout; this is the foundation of the site's visual cohesion
- **History timeline** — the best-designed interaction on the site; scroll animation, alternating layout, and dual-color system are executed with care
- **Fluid typography** — clamp() scaling means no jarring size jumps at breakpoints
- **Toast notification system** — context-aware colors (coral for favourites, teal for info) extend the design language into micro-interactions
- **Skeleton loading** — shimmer placeholders on gallery cards prevent layout shift, a sign of thoughtful UX
- **Glassmorphic nav** — backdrop-filter blur is a subtle, modern detail that works well with the cream background
- **Responsive gallery grid** — auto-fill + minmax() adjusts column count without media queries

Priority Recommendations (Remaining)

1	Add a tablet breakpoint (1024 px) The single 768 px breakpoint leaves tablet users in an in-between state; a second breakpoint would significantly improve the mid-size experience
2	Zero-results state for breed filter Currently the gallery goes blank with no message when a filter returns no cats; a friendly empty state prevents user confusion
3	Fix hamburger aria-expanded The toggle state is never communicated to assistive technology; a one-line JS fix in app.js resolves this for screen reader users
4	Add aria-hidden to decorative emoji All emoji used as icons should have aria-hidden='true' to prevent screen readers announcing them as literal emoji names
5	Consolidate page-level hero CSS into style.css Inline <style> blocks in each HTML file make maintenance harder; moving them to named sections in style.css keeps the design system in one place

Summary Score

Dimension	Rating	Notes
Visual Design	4 / 5	Strong palette, great typography, minor consistency gaps
Usability	4 / 5	Clear navigation and flows; a few rough edges in the gallery
Accessibility	3 / 5	Contrast and focus issues fixed; emoji and ARIA still need attention
Consistency	4 / 5	Excellent design system; a few one-off values to clean up
Mobile	3 / 5	Works, but a single breakpoint leaves tablet experience underserved
Overall	4 / 5	A polished, personality-rich site with a solid foundation

Report prepared by Claude Code — March 2026